

Intrinsic-Dimension-Dependent Wasserstein Bounds for Denoising Diffusion Probabilistic Models

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Abstract

We study the convergence of the denoising diffusion probabilistic model (DDPM) in the quadratic Wasserstein distance when the target distribution has low intrinsic complexity. The data distribution may be singular with respect to Lebesgue measure; its low-dimensional structure is quantified by a metric-entropy condition on its support. We represent the original DDPM update as the grid-point values of a semilinear stochastic differential equation and compare this interpolation with the exact reverse-time Ornstein–Uhlenbeck diffusion by a synchronous coupling. Tweedie’s formula rewrites the nonlinear drift in terms of the posterior mean of the clean data. Along the reverse process, this posterior mean is a square-integrable martingale, which yields an orthogonal decomposition between the propagated sampling error and the local time-discretization error. A posterior-covariance estimate inherited from the metric-entropy structure then converts the ambient-dimensional discretization term into a quantity of order $k \log k$, where k is the intrinsic dimension. The final non-asymptotic bound separates early-stopping, Gaussian-initialization, score-estimation, and time-discretization errors. In particular, the discretization component depends on the intrinsic dimension rather than directly on the ambient dimension. We also identify the remaining sources of ambient-dimensional dependence and the regularity assumptions needed by the coupling argument.

Keywords. Diffusion models; DDPM; Wasserstein distance; intrinsic dimension; metric entropy; reverse-time diffusion; Tweedie’s formula.

1 Introduction

Diffusion models generate samples from an unknown target distribution by reversing a gradual noising procedure. Their empirical performance in high-dimensional applications contrasts with worst-case convergence bounds whose complexity scales with the ambient dimension. This discrepancy has motivated a growing line of work on whether diffusion samplers automatically exploit low-dimensional structure in the data distribution.

The low-dimensional structure considered here is deliberately distribution-free: the support of the target law need not be a smooth manifold and the target law need not possess a density. Instead, we assume that the support admits a small Euclidean covering at a polynomially fine scale. Such a metric-entropy condition is broad enough to include many compact low-dimensional sets, while retaining the quantitative posterior-concentration estimates that drive recent intrinsic-dimension analyses of DDPMs.

Most sharp intrinsic-dimension results for DDPMs have been formulated in Kullback–Leibler divergence. Wasserstein analysis is complementary. The quadratic Wasserstein distance is sensitive to the geometry of the sample space, remains meaningful for singular target measures with a finite second moment, and directly controls errors of expectations for suitable test functions. Its analysis is nevertheless more delicate: the path-space change-of-measure argument

commonly used for KL divergence is not directly available, and one must instead control the distance between coupled trajectories.

This paper develops such a coupling analysis for the original DDPM coefficients. The argument combines four ingredients.

- (i) The exact reverse-time Ornstein–Uhlenbeck diffusion is synchronously coupled with a continuous-time interpolation of DDPM.
- (ii) Tweedie’s formula expresses the score through the posterior mean $\mu_t(x) = \mathbb{E}[X_0 \mid X_t = x]$.
- (iii) The process $\mu_{T-t}(Y_t)$ is a martingale in reverse time. This removes a potentially costly cross term and allows local discretization errors to be accumulated in quadratic mean.
- (iv) A posterior-covariance bound controlled by the support metric entropy yields an intrinsic-dimensional estimate of the temporal fluctuations of $\mu_{T-t}(Y_t)$.

The main theorem separates four error sources: early stopping, replacement of the noised terminal law by a standard Gaussian, score estimation, and time discretization. The discretization error is bounded by

$$C \left\{ \kappa \min(\mathbb{E} \|X_0\|_2^2, Tk \log k) + \kappa^2 Nk \log k \right\}^{1/2},$$

up to a stability factor determined by the Lipschitz modulus of the learned score. Thus the principal discretization term depends on $k \log k$, rather than directly on the ambient dimension d .

Relation to prior work. The DDPM sampler was introduced by Ho et al. [12], and its SDE formulation and reverse-time interpretation were developed systematically by Song et al. [16]; see also the classical time-reversal results of Anderson [1] and Haussmann and Pardoux [11]. Dimension-dependent non-asymptotic analyses under weak data assumptions include Benton et al. [4], Chen et al. [7, 8]. Adaptation to low-dimensional structure in KL divergence was investigated by Azangulov et al. [3], Huang et al. [13], Li and Yan [14], Potapchik et al. [15]. Wasserstein convergence of diffusion samplers has recently been studied under several regularity regimes, including log-concavity, semiconvexity, and learned-score regularity; see, for example, Arsenyan et al. [2], Beyler and Bach [5], Bruno and Sabanis [6], Gao et al. [10]. Our analysis imports the metric-entropy posterior-localization mechanism of Huang et al. [13] into a synchronous W_2 coupling.

Organization. Section 2 introduces the forward and reverse processes and the DDPM interpolation. Section 3 states the intrinsic-dimension and score assumptions. Section 4 gives the main theorem. The proof is developed in Section 5; technical estimates are collected in the appendices.

2 Diffusion model and continuous interpolation

2.1 Forward Ornstein–Uhlenbeck process

Let $X_0 \sim p_{\text{data}}$ be an $\mathbb{R}^d$ -valued random vector satisfying $\mathbb{E} \|X_0\|_2^2 < \infty$. Consider the Ornstein–Uhlenbeck process

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad t \in [0, T], \quad (1)$$

where W is a standard d -dimensional Brownian motion independent of X_0 . Its marginal law admits the representation

$$X_t \stackrel{d}{=} e^{-t} X_0 + \sigma_t Z, \quad \sigma_t := \sqrt{1 - e^{-2t}}, \quad Z \sim \mathcal{N}(0, I_d), \quad (2)$$

with Z independent of X_0 . For $t > 0$, let q_t be the density of X_t and define the score

$$s_t(x) := \nabla \log q_t(x). \quad (3)$$

2.2 Reverse-time diffusion

The reverse-time SDE associated with (1) is

$$dY_t = \{Y_t + 2s_{T-t}(Y_t)\} dt + \sqrt{2} dB_t, \quad t \in [0, T], \quad (4)$$

where B is a Brownian motion in the reverse filtration. If $Y_0 \sim \mathcal{L}(X_T)$, then

$$Y_t \stackrel{d}{=} X_{T-t}, \quad 0 \leq t \leq T. \quad (5)$$

For later use, define the posterior mean and posterior covariance

$$\mu_t(x) := \mathbb{E}[X_0 \mid X_t = x], \quad \Sigma_t(x) := \text{Cov}(X_0 \mid X_t = x). \quad (6)$$

Tweedie's formula [9] gives

$$\mu_t(x) = \frac{x + \sigma_t^2 s_t(x)}{e^{-t}}, \quad s_t(x) = \frac{e^{-t}}{\sigma_t^2} \mu_t(x) - \frac{x}{\sigma_t^2}. \quad (7)$$

Consequently, (4) can be rewritten as

$$dY_t = \left\{ \left(1 - \frac{2}{\sigma_{T-t}^2}\right) Y_t + \frac{2e^{-(T-t)}}{\sigma_{T-t}^2} \mu_{T-t}(Y_t) \right\} dt + \sqrt{2} dB_t. \quad (8)$$

2.3 DDPM and its semilinear interpolation

Choose grid points

$$0 = t_0 < t_1 < \dots < t_N < t_{N+1} = T, \quad h_n := t_{n+1} - t_n. \quad (9)$$

Set

$$\alpha_n := e^{-2h_n}, \quad \bar{\alpha}_n := \prod_{i=n}^N \alpha_i = e^{-2(T-t_n)}. \quad (10)$$

Given score estimators $\hat{s}_t$, the DDPM update is

$$\begin{aligned} \hat{Y}_{t_0} &\sim \mathcal{N}(0, I_d), \\ \hat{Y}_{t_{n+1}} &= \frac{1}{\sqrt{\alpha_n}} \left\{ \hat{Y}_{t_n} + (1 - \alpha_n) \hat{s}_{T-t_n}(\hat{Y}_{t_n}) \right\} + \sqrt{\frac{(1 - \alpha_n)(1 - \bar{\alpha}_{n+1})}{1 - \bar{\alpha}_n}} Z_n, \end{aligned} \quad (11)$$

where $Z_0, \dots, Z_{N-1}$ are independent standard Gaussian vectors.

Define the posterior-mean estimator induced by $\hat{s}_t$ as

$$\hat{\mu}_t(x) := \frac{x + \sigma_t^2 \hat{s}_t(x)}{e^{-t}}. \quad (12)$$

On each interval $[t_n, t_{n+1}]$, consider the semilinear SDE

$$d\hat{Y}_t = \left\{ \left(1 - \frac{2}{\sigma_{T-t}^2}\right) \hat{Y}_t + \frac{2e^{-(T-t)}}{\sigma_{T-t}^2} \hat{\mu}_{T-t_n}(\hat{Y}_{t_n}) \right\} dt + \sqrt{2} dB_t. \quad (13)$$

The linear part is integrated exactly, while the nonlinear posterior-mean term is frozen at the left endpoint.

Proposition 2.1 (Exact grid-point representation). *The grid-point values of (13) satisfy (11) in distribution. In particular, (13) is a continuous-time interpolation of the original DDPM sampler.*

Proof. Applying the variation-of-constants formula on $[t_n, t_{n+1}]$ gives

$$\begin{aligned} \hat{Y}_{t_{n+1}} = & e^{-h_n} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-t_n}^2} \hat{Y}_{t_n} + \frac{e^{-(T-t_{n+1})}(1-e^{-2h_n})}{\sigma_{T-t_n}^2} \hat{\mu}_{T-t_n}(\hat{Y}_{t_n}) \\ & + \sqrt{2} \int_{t_n}^{t_{n+1}} e^{-(t_{n+1}-u)} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-u}^2} dB_u. \end{aligned} \quad (14)$$

Substituting (12) and using

$$e^{-h_n} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-t_n}^2} + \frac{e^{h_n}(1-e^{-2h_n})}{\sigma_{T-t_n}^2} = e^{h_n}$$

produces the deterministic part of (11). A direct Itô-isometry calculation shows that the Gaussian integral in (14) has covariance

$$\frac{(1-\alpha_n)(1-\bar{\alpha}_{n+1})}{1-\bar{\alpha}_n} I_d.$$

Independence across disjoint time intervals completes the proof. $\square$

3 Intrinsic dimension and analytic assumptions

3.1 Metric-entropy structure

Definition 3.1 (Covering number). For $A \subset \mathbb{R}^d$ and $\varepsilon > 0$, let $N_{\text{cov}}(A, \|\cdot\|_2, \varepsilon)$ be the smallest integer m for which there exist $x_1, \dots, x_m \in \mathbb{R}^d$ such that

$$A \subset \bigcup_{i=1}^m B(x_i, \varepsilon).$$

Let $\mathcal{X} := \text{supp}(p_{\text{data}})$.

Assumption 3.2 (Metric-entropy dimension). *Let $k \geq 2$. There are fixed constants $C_0, C_{\text{cov}} > 0$ such that, with $\varepsilon_0 = k^{-C_0}$,*

$$\log N_{\text{cov}}(\mathcal{X}, \|\cdot\|_2, \varepsilon_0) \leq C_{\text{cov}} k \log(1/\varepsilon_0). \quad (15)$$

Assumption 3.3 (Bounded support). *For a fixed constant $C_R > 0$,*

$$\sup_{x \in \mathcal{X}} \|x\|_2 \leq R, \quad R \leq k^{C_R}. \quad (16)$$

The dimension k in Assumption 3.2 is a metric-entropy notion of intrinsic dimension. No manifold parametrization, smoothness, or absolute continuity of p_{data} is required.

3.2 Time mesh and score assumptions

Let the early-stopping time be

$$\delta := T - t_N. \quad (17)$$

Define the relative mesh size κ by requiring

$$h_n \leq \kappa \min\{1, T - t_n\}, \quad 0 \leq n \leq N - 1. \quad (18)$$

For notational simplicity, we assume that the grid contains $T - 1$, so that $t_M = T - 1$ for some $M \in \{0, \dots, N\}$. This restriction can be removed by splitting the interval containing $T - 1$ and modifying only universal constants.

Assumption 3.4 (Spatial regularity of the learned score). *There is a measurable function $L : [\delta, T] \rightarrow [0, \infty)$ such that*

$$\|\widehat{s}_r(x) - \widehat{s}_r(y)\|_2 \leq L(r) \|x - y\|_2, \quad r \in [\delta, T], \quad x, y \in \mathbb{R}^d. \quad (19)$$

Assumption 3.5 (Score estimation along the forward marginals). *For $n = 0, \dots, N-1$, there are finite numbers $\varepsilon_n \geq 0$ such that*

$$\mathbb{E} \|s_{T-t_n}(X_{T-t_n}) - \widehat{s}_{T-t_n}(X_{T-t_n})\|_2^2 \leq \varepsilon_n^2. \quad (20)$$

4 Main result

Theorem 4.1 (Intrinsic-dimension-dependent Wasserstein bound). *Suppose Assumptions 3.2–3.5 hold. Assume $T > 1$, $k \geq 2$, and that for fixed constants $C_d, C_\delta > 0$,*

$$d \leq k^{C_d}, \quad k^{-C_\delta} \leq \delta < 1, \quad \kappa \leq 0.9. \quad (21)$$

Then there is a constant $C > 0$, depending only on the fixed constants in the assumptions, such that

$$\begin{aligned} W_2(p_{\text{data}}, \mathcal{L}(\widehat{Y}_{t_N})) &\leq \underbrace{\left\{ (1 - e^{-\delta})^2 \mathbb{E} \|X_0\|_2^2 + \sigma_\delta^2 d \right\}^{1/2}}_{\text{early-stopping error}} + \exp \left\{ t_N + 2 \sum_{i=0}^{N-1} h_i L(T - t_i) \right\} \underbrace{\left[\left\{ e^{-2T} \mathbb{E} \|X_0\|_2^2 + (1 - \sigma_T)^2 d \right\}^{1/2} \right]}_{\text{Gaussian-initialization error}} \\ &\quad + \underbrace{\sum_{j=0}^{N-1} e^{h_j} (1 - e^{-2h_j}) \varepsilon_j}_{\text{score-estimation error}} + \underbrace{C \left\{ \kappa \min(\mathbb{E} \|X_0\|_2^2, Tk \log k) + \kappa^2 Nk \log k \right\}^{1/2}}_{\text{time-discretization error}}. \end{aligned} \quad (22)$$

Moreover,

$$\mathcal{G}_N \leq \exp \left\{ t_N + 2 \sum_{i=0}^{N-1} h_i L(T - t_i) \right\}. \quad (23)$$

Remark 4.2 (Where the ambient dimension remains). *The principal discretization term in (22) is governed by $k \log k$. The ambient dimension d remains in the early-stopping and Gaussian-initialization errors. These terms arise before the intrinsic-dimensional posterior fluctuation estimate is used. Choosing T large and δ small controls them, but the required choices may depend on d .*

Remark 4.3 (Score regularity). *The Lipschitz assumption is imposed only on the learned score, not on the target density or the true score. It enters through the pathwise stability of the synchronous coupling. Removing this assumption would require a different stability mechanism, such as one-sided contractivity or a localized argument.*

5 Proof of the main theorem

5.1 A coupling compatible with the reverse filtration

A subtle point is that the martingale argument below must remain valid after the approximate process is introduced. We therefore use a coupling that preserves the conditional structure of the reverse process.

Let $(X_0, (Y_t)_{0 \leq t \leq t_N})$ be a realization of the exact reverse process satisfying (5). Couple $Y_0 \sim \mathcal{L}(X_T)$ and $\hat{Y}_0 \sim \mathcal{N}(0, I_d)$ through an optimal W_2 coupling. Equivalently, conditionally on Y_0 , generate $\hat{Y}_0$ from a Markov kernel that realizes an optimal transport plan. The auxiliary randomness used by this kernel is conditionally independent of X_0 given Y_0 . Starting from $(Y_0, \hat{Y}_0)$, drive (8) and (13) with the same Brownian motion.

Let

$$\mathcal{F}_t := \sigma\{Y_s, \hat{Y}_s : 0 \leq s \leq t\}.$$

After the usual completion, because $\hat{Y}_0$ reveals no additional information about X_0 once Y_0 is known, and because $\hat{Y}$ is generated from $(\hat{Y}_0, B)$, the reverse Markov property gives

$$\mathbb{E}[X_0 \mid \mathcal{F}_t] = \mu_{T-t}(Y_t). \quad (24)$$

This construction avoids enlarging the filtration by a noise variable that would reveal extra information about X_0 .

The initialization distance is bounded independently by the explicit marginal coupling in (2):

$$\begin{aligned} W_2^2(\mathcal{L}(X_T), \mathcal{N}(0, I_d)) &\leq \mathbb{E} \|e^{-T} X_0 + (\sigma_T - 1)Z\|_2^2 \\ &= e^{-2T} \mathbb{E} \|X_0\|_2^2 + (1 - \sigma_T)^2 d. \end{aligned} \quad (25)$$

5.2 Posterior-mean martingale

Lemma 5.1 (Reverse posterior martingale). *The process*

$$M_t := \mu_{T-t}(Y_t), \quad 0 \leq t \leq t_N, \quad (26)$$

is a square-integrable $\mathcal{F}_t$ -martingale. Moreover, for $0 \leq s \leq t \leq t_N$,

$$\mathbb{E} \|M_t - M_s\|_2^2 = \mathbb{E} \text{Tr} \Sigma_{T-s}(Y_s) - \mathbb{E} \text{Tr} \Sigma_{T-t}(Y_t). \quad (27)$$

Proof. Equation (24) proves the martingale property. For $s \leq t$, the orthogonality of conditional expectations yields

$$\mathbb{E} \|X_0 - M_s\|_2^2 = \mathbb{E} \|X_0 - M_t\|_2^2 + \mathbb{E} \|M_t - M_s\|_2^2.$$

Since

$$\mathbb{E} \|X_0 - M_r\|_2^2 = \mathbb{E} \text{Tr} \Sigma_{T-r}(Y_r),$$

we obtain (27). □

5.3 One-step error decomposition

Let

$$\Delta_n := Y_{t_n} - \hat{Y}_{t_n}.$$

The exact reverse process satisfies

$$\begin{aligned} Y_{t_{n+1}} &= e^{-h_n} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-t_n}^2} Y_{t_n} \\ &\quad + \int_{t_n}^{t_{n+1}} e^{-(t_{n+1}-u)} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-u}^2} \frac{2e^{-(T-u)}}{\sigma_{T-u}^2} \mu_{T-u}(Y_u) du \\ &\quad + \sqrt{2} \int_{t_n}^{t_{n+1}} e^{-(t_{n+1}-u)} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-u}^2} dB_u. \end{aligned} \quad (28)$$

A direct calculation gives

$$\int_{t_n}^{t_{n+1}} e^{-(t_{n+1}-u)} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-u}^2} \frac{2e^{-(T-u)}}{\sigma_{T-u}^2} du = \frac{e^{-(T-t_{n+1})}(1-e^{-2h_n})}{\sigma_{T-t_n}^2}. \quad (29)$$

Define the local remainder

$$R_n := \int_{t_n}^{t_{n+1}} e^{-(t_{n+1}-u)} \frac{\sigma_{T-t_{n+1}}^2}{\sigma_{T-u}^2} \frac{2e^{-(T-u)}}{\sigma_{T-u}^2} \times \{\mu_{T-u}(Y_u) - \mu_{T-t_n}(Y_{t_n})\} du. \quad (30)$$

Subtracting (14) from (28), using (7)–(12), yields

$$\begin{aligned} \Delta_{n+1} &= e^{h_n} \Delta_n \\ &\quad + e^{h_n}(1-e^{-2h_n})\{s_{T-t_n}(Y_{t_n}) - \widehat{s}_{T-t_n}(Y_{t_n})\} \\ &\quad + e^{h_n}(1-e^{-2h_n})\{\widehat{s}_{T-t_n}(Y_{t_n}) - \widehat{s}_{T-t_n}(\widehat{Y}_{t_n})\} + R_n. \end{aligned} \quad (31)$$

Let S_n denote the first three terms on the right-hand side. Since M_t is a martingale and the kernel in (30) is deterministic,

$$\mathbb{E}[R_n \mid \mathcal{F}_{t_n}] = 0. \quad (32)$$

Because S_n is $\mathcal{F}_{t_n}$ -measurable,

$$\mathbb{E} \|\Delta_{n+1}\|_2^2 = \mathbb{E} \|S_n\|_2^2 + \mathbb{E} \|R_n\|_2^2. \quad (33)$$

By Minkowski's inequality, Assumptions 3.4 and 3.5, and $Y_{t_n} \stackrel{d}{=} X_{T-t_n}$,

$$\begin{aligned} (\mathbb{E} \|S_n\|_2^2)^{1/2} &\leq e^{h_n} \{1 + L(T-t_n)(1-e^{-2h_n})\} (\mathbb{E} \|\Delta_n\|_2^2)^{1/2} \\ &\quad + e^{h_n}(1-e^{-2h_n})\varepsilon_n. \end{aligned} \quad (34)$$

5.4 A discrete quadratic recursion

Lemma 5.2 (Quadratic discrete Gronwall inequality). *Let $x_n, B_n, C_n \geq 0$ and $A_n \in \mathbb{R}$ satisfy*

$$x_{n+1}^2 \leq (e^{A_n} x_n + B_n)^2 + C_n^2. \quad (35)$$

Then

$$\begin{aligned} x_{n+1} &\leq e^{\sum_{i=0}^n A_i} x_0 + \sum_{j=0}^n e^{\sum_{i=j+1}^n A_i} B_j \\ &\quad + \left\{ \sum_{j=0}^n e^{2\sum_{i=j+1}^n A_i} C_j^2 \right\}^{1/2}. \end{aligned} \quad (36)$$

Proof. View the right-hand side of (35) as the squared Euclidean norm of $(e^{A_n} x_n + B_n, C_n) \in \mathbb{R}^2$ and iterate Minkowski's inequality. $\square$

Applying Lemma 5.2 to (33)–(34) and using that each stability factor is at least one gives

$$\begin{aligned} (\mathbb{E} \|\Delta_N\|_2^2)^{1/2} &\leq \mathcal{G}_N \left[(\mathbb{E} \|\Delta_0\|_2^2)^{1/2} + \sum_{j=0}^{N-1} e^{h_j}(1-e^{-2h_j})\varepsilon_j \right. \\ &\quad \left. + \left\{ \sum_{j=0}^{N-1} \mathbb{E} \|R_j\|_2^2 \right\}^{1/2} \right]. \end{aligned} \quad (37)$$

By the optimal initial coupling and (25),

$$\mathbb{E} \|\Delta_0\|_2^2 \leq e^{-2T} \mathbb{E} \|X_0\|_2^2 + (1-\sigma_T)^2 d. \quad (38)$$

5.5 Intrinsic-dimensional posterior covariance

The key geometric input is the following estimate.

Lemma 5.3 (Posterior covariance bound). *Under Assumptions 3.2 and 3.3 and (21), there is a constant $C_\Sigma > 0$ such that, for $0 \leq t \leq t_N$,*

$$\mathbb{E} \operatorname{Tr} \Sigma_{T-t}(Y_t) \leq C_\Sigma \min \left\{ \mathbb{E} \|X_0\|_2^2, \frac{\sigma_{T-t}^2}{1 - \sigma_{T-t}^2} k \log k \right\}. \quad (39)$$

A proof is given in Section A. The first bound follows from the least-squares optimality of conditional expectation. The second is obtained by covering $\mathcal{X}$ by an ε_0 -net, discarding cells of exponentially small mass, and applying Gaussian concentration to localize the posterior around the net point compatible with the noisy observation. The polynomial assumptions on d , R , and δ^{-1} absorb the exponentially small exceptional terms.

5.6 Discretization remainder

Lemma 5.4 (Accumulated local remainder). *Under the assumptions of Theorem 4.1,*

$$\sum_{n=0}^{N-1} \mathbb{E} \|R_n\|_2^2 \leq C \left\{ \kappa \min(\mathbb{E} \|X_0\|_2^2, Tk \log k) + \kappa^2 N k \log k \right\}. \quad (40)$$

Proof. By Cauchy–Schwarz, the mesh condition, and elementary bounds on the deterministic kernel in (30),

$$\mathbb{E} \|R_n\|_2^2 \leq 4 \int_{t_n}^{t_{n+1}} \frac{1 - \sigma_{T-u}^2}{\sigma_{T-u}^4} \mathbb{E} \|M_u - M_{t_n}\|_2^2 du. \quad (41)$$

Using (27), define

$$V(t) := \mathbb{E} \operatorname{Tr} \Sigma_{T-t}(Y_t).$$

Then V is nonincreasing and

$$\sum_{n=0}^{N-1} \mathbb{E} \|R_n\|_2^2 \leq 4 \sum_{n=0}^{N-1} \int_{t_n}^{t_{n+1}} \frac{1 - \sigma_{T-u}^2}{\sigma_{T-u}^4} \{V(t_n) - V(u)\} du. \quad (42)$$

We split the sum at $t_M = T - 1$. For $u \leq T - 1$, the denominator is bounded away from zero. Discrete summation by parts, the monotonicity of V , and the two alternatives in (39) give

$$\begin{aligned} \sum_{n=0}^{M-1} \int_{t_n}^{t_{n+1}} \frac{1 - \sigma_{T-u}^2}{\sigma_{T-u}^4} \{V(t_n) - V(u)\} du \\ \leq C \kappa \min \left\{ \mathbb{E} \|X_0\|_2^2, Tk \log k \right\}. \end{aligned} \quad (43)$$

For $T - 1 \leq u \leq T - \delta$,

$$\frac{1 - \sigma_{T-u}^2}{\sigma_{T-u}^4} = \frac{e^{-2(T-u)}}{(1 - e^{-2(T-u)})^2} \leq \frac{C}{(T - u)^2}. \quad (44)$$

Moreover, (18) implies

$$h_n \leq \frac{\kappa}{1 - \kappa} (T - t_{n+1}). \quad (45)$$

A second discrete summation-by-parts argument therefore yields

$$\begin{aligned} & \sum_{n=M}^{N-1} \int_{t_n}^{t_{n+1}} \frac{1 - \sigma_{T-u}^2}{\sigma_{T-u}^4} \{V(t_n) - V(u)\} du \\ & \leq C\kappa V(T-1) + C\kappa^2 \sum_{n=M+1}^{N-1} \frac{V(t_n)}{T-t_n}. \end{aligned} \quad (46)$$

By (39) and

$$\frac{1}{r} \frac{\sigma_r^2}{1 - \sigma_r^2} = \frac{e^{2r} - 1}{r} \leq 2e^2, \quad 0 < r \leq 1,$$

we obtain

$$(46) \leq C\kappa \min\{\mathbb{E} \|X_0\|_2^2, k \log k\} + C\kappa^2 N k \log k. \quad (47)$$

Combining (42), (43), and (47) proves the claim. $\square$

5.7 Conclusion of the proof

Since $Y_{t_N} \stackrel{d}{=} X_\delta$,

$$W_2(\mathcal{L}(Y_{t_N}), \mathcal{L}(\hat{Y}_{t_N})) \leq (\mathbb{E} \|\Delta_N\|_2^2)^{1/2}. \quad (48)$$

The explicit forward coupling gives

$$\begin{aligned} W_2^2(p_{\text{data}}, \mathcal{L}(X_\delta)) & \leq \mathbb{E} \left\| X_0 - e^{-\delta} X_0 - \sigma_\delta Z \right\|_2^2 \\ & = (1 - e^{-\delta})^2 \mathbb{E} \|X_0\|_2^2 + \sigma_\delta^2 d. \end{aligned} \quad (49)$$

The triangle inequality, (37), (38), and Lemma 5.4 yield (22). Finally, $1 + x \leq e^x$ and $1 - e^{-2h_i} \leq 2h_i$ imply

$$\mathcal{G}_N \leq \exp \left\{ \sum_{i=0}^{N-1} h_i + 2 \sum_{i=0}^{N-1} h_i L(T - t_i) \right\},$$

which is (23). This completes the proof of Theorem 4.1.

6 Discussion

Why the posterior mean is the right state variable. A direct discretization estimate for the score can introduce derivatives of the score and ambient-dimensional moment terms. Tweedie's formula replaces the nonlinear score contribution by the posterior mean. Its reverse-time martingale structure then turns temporal increments into differences of posterior variances, which can be bounded by intrinsic geometry.

Role of the exponential integrator. The interpolation (13) exactly reproduces the DDPM coefficients at grid points. This exact correspondence is useful for interpreting the theorem as a guarantee for the original DDPM sampler rather than for a nearby Euler–Maruyama scheme. A standard Euler discretization can also be analyzed, but it produces additional local errors from the singular linear coefficient near the data endpoint and no longer matches (11) exactly.

Limitations. The theorem requires bounded support, a one-scale metric-entropy condition, a polynomial relation between d and k , and a Lipschitz learned score. The stability factor may be large when the learned score has a large Lipschitz modulus. In addition, ambient dimension remains in the initialization and early-stopping terms. Obtaining a completely intrinsic-dimensional W_2 bound under minimal score regularity remains open.

Possible extensions. Natural directions include replacing global Lipschitz continuity by one-sided or average regularity, extending the posterior-covariance argument to unbounded distributions with concentration assumptions, optimizing the nonuniform time mesh, and analyzing the robustness of the intrinsic-dimensional bound under perturbations of the DDPM coefficients.

A Posterior covariance under metric entropy

We give the argument underlying Lemma 5.3. Let $r = T - t \in [\delta, T]$. By (5),

$$\mathbb{E} \operatorname{Tr} \Sigma_{T-t}(Y_t) = \mathbb{E} \operatorname{Tr} \Sigma_r(X_r). \quad (50)$$

The least-squares property of conditional expectation immediately gives

$$\mathbb{E} \operatorname{Tr} \Sigma_r(X_r) = \mathbb{E} \|X_0 - \mathbb{E}[X_0 \mid X_r]\|_2^2 \leq \mathbb{E} \|X_0\|_2^2. \quad (51)$$

For the intrinsic-dimensional estimate, let $\{x_i^*\}_{i=1}^m$ be an ε_0 -net of $\mathcal{X}$. By Assumption 3.2,

$$m \leq \exp(Ck \log k). \quad (52)$$

Partition $\mathcal{X}$ into measurable sets $B_i \subset B(x_i^*, \varepsilon_0)$. Retain the cells whose masses are at least $\exp(-C_6 k \log k)$, where C_6 is a sufficiently large constant. The total mass of discarded cells is at most $\exp(-cC_6 k \log k)$.

Write $X_r = e^{-r} X_0 + \sigma_r Z$. A union bound combined with Gaussian concentration produces an event $\mathcal{G}$ with probability at least $1 - \exp(-cC_6 k \log k)$ on which all pairwise Gaussian inner products associated with the net are bounded by $C\sqrt{k \log k}$ times the corresponding net distance. The standard Bayes-ratio comparison used in metric-entropy analyses of DDPMs then implies that, for every typical observation x , the conditional law of X_0 given $X_r = x$ is concentrated around a compatible net point $x_{i(x)}^*$ at radius

$$C \frac{\sigma_r}{e^{-r}} \sqrt{k \log k} = C \left(\frac{\sigma_r^2}{1 - \sigma_r^2} k \log k \right)^{1/2}, \quad (53)$$

up to conditional probability $\exp(-ck \log k)$. This posterior-localization statement is precisely the covering argument developed for the low-dimensional DDPM analysis of Huang et al. [13].

By conditional least-squares optimality,

$$\begin{aligned} \operatorname{Tr} \Sigma_r(x) &\leq \mathbb{E} [\|X_0 - x_{i(x)}^*\|_2^2 \mid X_r = x] \\ &\leq C \frac{\sigma_r^2}{1 - \sigma_r^2} k \log k + 4R^2 e^{-ck \log k} \end{aligned} \quad (54)$$

for typical x , while the exceptional observations contribute at most $R^2 e^{-ck \log k}$. Hence

$$\mathbb{E} \operatorname{Tr} \Sigma_r(X_r) \leq C \frac{\sigma_r^2}{1 - \sigma_r^2} k \log k + CR^2 e^{-ck \log k}. \quad (55)$$

Because $r \geq \delta \geq k^{-C_\delta}$,

$$\frac{\sigma_r^2}{1 - \sigma_r^2} = e^{2r} - 1 \geq c\delta \geq ck^{-C_\delta}.$$

Together with $R \leq k^{C_R}$ and $d \leq k^{C_d}$, this polynomial lower bound absorbs the exponentially small term in (55) after increasing the universal constants. Combining the result with (51) proves (39).

B A summation-by-parts identity

For completeness, we record the elementary identity used in the remainder analysis. Let $a_0, \dots, a_m \geq 0$ and let $v_0 \geq v_1 \geq \dots \geq v_{m+1} \geq 0$. Then

$$\sum_{j=0}^m a_j(v_j - v_{j+1}) = a_0 v_0 - a_m v_{m+1} + \sum_{j=1}^m (a_j - a_{j-1}) v_j. \quad (56)$$

Upper bounds follow by controlling the positive increments of a_j . In Lemma 5.4, v_j is a posterior-variance sequence and a_j is obtained by integrating the deterministic singular kernel over the j th time interval.

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